

Supplemental Material for Interlayer-distance control of spectral reconstruction in twisted graphene

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SCOPE AND PROVENANCE

This supplement documents the first-principles spectra, their underlying calculation and plotting conventions, and the numerical reproduction of the supplied tight-binding model. The July 23, 2026 data package contains ten SZV DOS maps, three DZVP maps, 51 input configurations at $d = 2.15 \text{ \AA}$, and model notebooks. The additional September 16, 2026 package supplies CP2K inputs, outputs, state-resolved projections, structure-generation and plotting notebooks, and seven monolayer DOS files. Table S2 records the usable state-resolved data for the original image series. In particular, SZV output at $d = 2.15 \text{ \AA}$ is absent from the additional package, although its input structures and original image are available.

The bilayer DFT panels retain the original images. The additional data identify the plotted component, spin channel, energy conversion, broadening, and column normalization. A monolayer reference figure uses the supplied DOS files with a common display convention. Public spectral extracts preserve the printed eigenvalues, occupations, and lower-layer weights, while a machine-readable inventory records input/output hashes, convergence flags, and incomplete files.

The coordinate audit identifies a naming discrepancy: the directory and original image labeled $2.85 \text{ \AA}$ contain a planar separation of $2.84 \text{ \AA}$ in all 51 inputs, with cell height $12.84 \text{ \AA}$. The figure label and table therefore use $2.84 \text{ \AA}$, while source filenames remain unchanged for traceability. The ancillary $3.35 \text{ \AA}$ series is retained in the data inventory but is not added to the ten-distance image survey.

ATOMISTIC GEOMETRY AND ITS LIMITATIONS

The supplied $d = 2.15 \text{ \AA}$ series samples $\theta = 30^\circ, 30.6^\circ, \dots, 60^\circ$. The cell vectors, in angstroms, are

$$\mathbf{A} = (61.4878, 0, 0), \quad \mathbf{B} = (30.7439, 53.25, 0), \quad \mathbf{C} = (0, 0, 12.15). \quad (\text{S1})$$

The first 1250 atoms are carbon in the lower layer at $z = 0$. They form a 25×25 supercell of the two-atom graphene primitive cell. The nearest-neighbor distance is $a_{\text{CC}} = 1.42 \text{ \AA}$. The primitive-vector length is $\sqrt{3}a_{\text{CC}}$, not a_{CC} . The upper sheet is rotated and cut to avoid overlap with its periodic replicas, and its dangling bonds are hydrogen terminated. The supplied C–H distances are approximately $1.09 \text{ \AA}$. All upper carbon and hydrogen atoms lie at $z = 2.15 \text{ \AA}$.

The finite upper patch contains 955–1013 carbon atoms and 84–100 hydrogen atoms. The total atom count ranges from 2290 to 2362. For example, the 45° structure has 984 upper-layer carbon atoms and 94 hydrogen atoms, giving 2328 atoms in total. Figure S1 and the accompanying `data/geometry_counts.csv` record this variation. An angle scan consequently changes not only registry but also coverage and edge termination. These variations are not equivalent to a twist scan at fixed Hilbert-space dimension in the disk model.

The input specifies periodicity in all three directions. At this separation, the distance from the upper plane to the next lower-layer periodic image along z is $10 \text{ \AA}$. An isolated-boundary Poisson block is commented out. These facts do not establish convergence with respect to vacuum or electrostatic boundary conditions. The calculation type is **ENERGY**: neither structural relaxation nor force convergence is demonstrated. An imposed layer separation is not a measured or calculated hydrostatic pressure. In particular, the strongly compressed structures are constrained electronic configurations. Their mechanical stability remains to be tested.

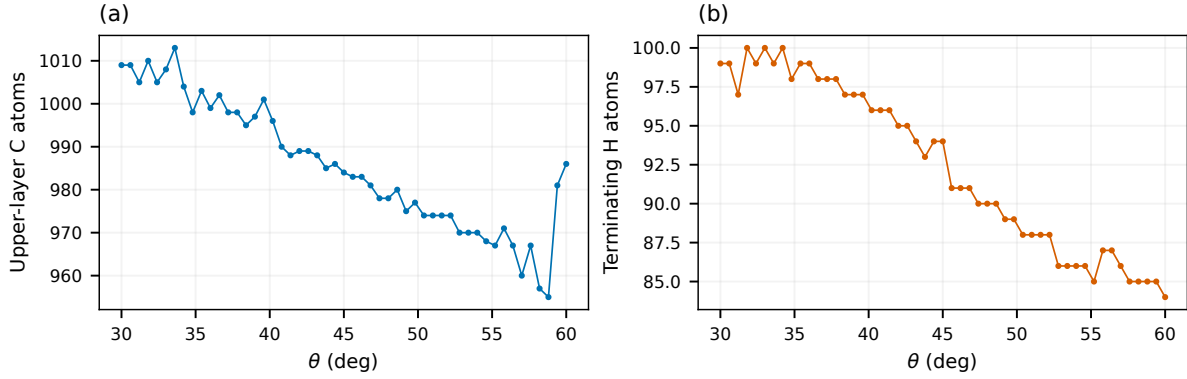


FIG. S1. Atom counts determined directly from the 51 supplied coordinate files at $d = 2.15 \text{ \AA}$. The lower layer contains 1250 carbon atoms at every angle. Upper-layer carbon and terminating hydrogen counts vary with the construction of the finite patch.

FIRST-PRINCIPLES SETTINGS AND PROJECTED SPECTRAL WEIGHTS

Table S1 summarizes the settings of the ten-distance survey recovered from the supplied inputs and output headers. The latter identify CP2K 2023.1, except for the supplied $d = 3.30 \text{ \AA}$, $\theta = 49.8^\circ$ record, which reports CP2K 2026.1 (development version) with the same listed settings. The relative density cutoff is 20 hartree, equivalent to 40 Ry. The basis identifiers are SZV-MOLOPT-GTH and DZVP-MOLOPT-GTH for both carbon and hydrogen. Qualitative similarity between their spectra is a basis comparison, not a complete basis-convergence test. The separate $3.35 \text{ \AA}$ reference series uses other run settings and versions, recorded in the inventory, and is not pooled with the survey.

The inputs specify the libraries `BASIS_MOLOPT` and `GTH_POTENTIALS`, not the newer UZH parameter files. The UZH protocol [1] provides the methodological reference for constructing MOLOPT/GTH parameter sets and separately assessing Gaussian-basis and pseudopotential errors.

TABLE S1. First-principles settings established from the supplied input and output files.

Electronic-structure method	CP2K Quickstep, primarily version 2023.1 (exception noted above), Gaussian and plane waves (GPW) [2, 3], PBE [4]
Pseudopotentials	GTH-PBE, carbon $q4$, hydrogen $q1$
Gaussian basis	SZV-MOLOPT-GTH, or DZVP-MOLOPT-GTH for the comparison series, for C and H
Spin and charge	Spin-polarized, neutral. Multiplicity not explicitly set in the input. Output multiplicities are 1 or 2, depending on electron-number parity
Density grid	300-Ry cutoff, 40-Ry relative cutoff, and five multigrids
Quickstep accuracy	<code>EPS_DEFAULT</code> = 10^{-10}
Self-consistency	<code>EPS_SCF</code> = 10^{-5} , maximum 6000 iterations. Atomic initial guess, with restart guesses in some inputs
Diagonalization and mixing	Standard diagonalization with adaptive threshold 0.01. Broyden mixing with $\alpha = 0.2$, buffer 10
Occupation scheme	Fermi-Dirac smearing at 300 K with 300 added molecular orbitals
Dispersion	D3 pair-potential input with C9 terms requested and long-range correction disabled [5]
Reciprocal-space sampling	Γ -point calculation (no explicit <code>KPOINTS</code> section)
DOS output request	<code>DELTA_E</code> = 0.0001 in the DOS print block
Layer projection	PDOS/LDOS on atoms 1–1250 with component output requested
Boundary and run type	Fixed-geometry <code>ENERGY</code> run with cell periodicity XYZ

The D3 term accounts for dispersion interactions that are important for interlayer binding in graphene and for its equilibrium layer separation [6]. It depends on the geometry and is added to the DFT energy and forces. It does not

by itself alter the Kohn–Sham potential at a fixed geometry. Dispersion could affect the spectrum indirectly through a relaxed structure, but such relaxation is not part of the supplied input series.

Let $\{\chi_\mu\}$ be the nonorthogonal Gaussian basis. The matrices

$$H_{\mu\nu,\sigma} = \langle \chi_\mu | \hat{H}_{\text{KS},\sigma} | \chi_\nu \rangle, \quad S_{\mu\nu} = \langle \chi_\mu | \chi_\nu \rangle \quad (\text{S2})$$

represent the Kohn–Sham operator and the basis overlap. They satisfy

$$\mathbf{H}_\sigma \mathbf{c}_{n\sigma} = \varepsilon_{n\sigma} \mathbf{S} \mathbf{c}_{n\sigma}, \quad \mathbf{c}_{n\sigma}^\dagger \mathbf{S} \mathbf{c}_{m\sigma} = \delta_{nm}. \quad (\text{S3})$$

For a positive-definite overlap matrix, symmetric orthogonalization [7] gives

$$\widetilde{\mathbf{H}}_\sigma = \mathbf{S}^{-1/2} \mathbf{H}_\sigma \mathbf{S}^{-1/2}, \quad \widetilde{\mathbf{c}}_{n\sigma} = \mathbf{S}^{1/2} \mathbf{c}_{n\sigma}. \quad (\text{S4})$$

The CP2K 2023.1 PDOS implementation uses the corresponding Löwdin-type coefficients. If $\mathbf{\Pi}_A$ selects a set A of orthogonalized orbital labels, then

$$w_{n\sigma}^{(A)} = \widetilde{\mathbf{c}}_{n\sigma}^\dagger \mathbf{\Pi}_A \widetilde{\mathbf{c}}_{n\sigma}, \quad 0 \leq w_{n\sigma}^{(A)} \leq 1. \quad (\text{S5})$$

For the bilayer images, $A = (L, p_z)$ contains only the p_z components on lower-layer atoms 1–1250. With the layers parallel to the xy plane, z is their common normal. The plotting notebooks read the alpha-spin `list1` file and select the sixth numerical column, whose header is `pz`. They neither sum all orbital components nor add the beta-spin channel. This is an orbital-partition weight, not a real-space slab integral. A complete partition of all orbital labels recovers unit weight for each normalized state, whereas the selected p_z contribution does not. Suppressed plotted weight can therefore coexist with eigenvalues of other orbital, layer, or spin character.

A Γ -point supercell spectrum consists of discrete eigenvalues. Broadening it is not equivalent to a converged Brillouin-zone integration. In an unrotated $N \times N$ graphene cell, primitive-cell K folds to the supercell Γ only if N is divisible by three. The present $N = 25$ does not satisfy that condition. A Dirac cone therefore cannot be diagnosed from a Γ -only angle–DOS map. Its horizontal coordinate is twist angle, not wave vector.

Energy scale, Gaussian kernel, and intensity normalization

The supplied SZV and DZVP plotting notebooks convert the printed Hartree energies using

$$e_{n\alpha} = 27.211384 [\varepsilon_{n\alpha} - E_F] \quad \text{in eV}, \quad (\text{S6})$$

where E_F is read from each PDOS header. They retain states with $-7 \leq e_{n\alpha} \leq 3$ eV and evaluate their weighted Gaussian sum on 6000 points,

$$E_j = \left(-7 + j \frac{10}{6000} \right) \text{eV}, \quad j = 0, \dots, 5999. \quad (\text{S7})$$

The display kernel used in the scripts is

$$K_\eta(x) = \frac{e^{-x^2/\eta^2}}{2\pi\eta}, \quad \eta = 0.02 \text{ eV}. \quad (\text{S8})$$

Its standard deviation is $\eta/\sqrt{2} = 0.01414$ eV and its full width at half maximum is $2\sqrt{\ln 2} \eta = 0.03330$ eV. The prefactor differs by the constant $1/(2\sqrt{\pi})$ from a unit-integral Gaussian. That constant cancels because each angle is displayed as

$$I(E_j; \theta) = \frac{\sum_{n \in \mathcal{W}_\alpha} w_{n\alpha}^{(L, p_z)} K_\eta(E_j - e_{n\alpha})}{\max_j \sum_{n \in \mathcal{W}_\alpha} w_{n\alpha}^{(L, p_z)} K_\eta(E_j - e_{n\alpha})}. \quad (\text{S9})$$

The unit-integral kernel in the main-text equation gives the same I . No occupation factor multiplies the weights. The 300-K Fermi–Dirac smearing sets the SCF occupations, not this Gaussian width. The original maps use alpha spin only and individual column maxima, so neither spin summation nor absolute intensity comparison between angles is implicit in their colors. The independently chosen Fermi origins also remove information about absolute electrostatic energy shifts.

Available states and completeness checks

The audit checks SCF convergence, normal program termination, and agreement between the number of printed PDOS rows and the number of molecular orbitals stated in the corresponding output. These conditions are evaluated separately for each spin. Some supplied PDOS files are truncated even when the output reports a converged calculation, so convergence alone does not validate an image column. The inventory retains these records with explicit flags rather than filling missing spectra by interpolation.

TABLE S2. State-resolved coverage of the original bilayer image series over 30° – 60° . An audited alpha column has a converged, normally terminated output and the expected number of PDOS rows. The final column gives the minimum and maximum, over these angles, of the highest printed alpha-spin energy relative to its own E_F . The images extend to +3 eV, but their numerical state coverage can end earlier.

Basis	d (Å)	Audited α columns / 51	Printed upper energy (eV)
SZV	3.30	50	2.71–2.83
SZV	3.00	51	2.75–2.90
SZV	2.84	51	2.79–2.93
SZV	2.70	51	2.84–2.93
SZV	2.55	51	2.84–2.99
SZV	2.30	48	2.78–3.07
SZV	2.25	46	2.76–3.05
SZV	2.20	44	2.74–2.98
SZV	2.15	not supplied	—
SZV	2.00	49	2.57–2.98
DZVP	2.25	49	2.36–2.56
DZVP	2.20	49	2.32–2.54
DZVP	2.15	47	2.29–2.60

The 300 added orbitals produce an angle- and basis-dependent upper energy limit. In particular, the DZVP data extend only to approximately 2.3–2.6 eV above E_F . Low intensity above a configuration’s last printed state is a truncation effect and is excluded from physical interpretation. Analysis near either end of the selected energy window must also allow for the Gaussian width. The public extracts store both spin channels separately, including occupations, Fermi levels, p_z weights, and summed lower-layer orbital weights. They permit reconstruction of the available columns without recovering numbers from image colors.

TIGHT-BINDING MODEL AND THE SMALL-ENERGY-SCALE ISSUE

The model starts from a honeycomb point set

$$P_R = \{n_1 \mathbf{a}_1 + n_2 \mathbf{a}_2 + \mathbf{b}_s : |n_1 \mathbf{a}_1 + n_2 \mathbf{a}_2 + \mathbf{b}_s| \leq R\}, \quad (\text{S10})$$

where $\mathbf{a}_1 = (\sqrt{3}a, 0)$, $\mathbf{a}_2 = (\sqrt{3}a/2, 3a/2)$, and $\mathbf{b}_s \in \{(0, 0), (0, a)\}$. The two-layer set is

$$X_{\theta,d} = \{(\mathbf{r}, 0) : \mathbf{r} \in P_R\} \cup \{(\mathbf{R}_\theta \mathbf{r}, d) : \mathbf{r} \in P_R\}. \quad (\text{S11})$$

Here $\mathbf{R}_\theta$ is a coordinate rotation matrix, not a Hilbert-space Hamiltonian. The operator $\hat{H}$ acts on orthonormal site states. Its matrix elements are

$$H_{ij} = \begin{cases} t_0 \exp(-|\mathbf{x}_i - \mathbf{x}_j|/\xi), & i \neq j, \\ 0, & i = j. \end{cases} \quad (\text{S12})$$

All pairs are retained. The supplied notebook uses $t_0 = 1$ in arbitrary energy units, with positive real hopping. There is no magnetic Peierls phase, orbital-angular factor, or fit to a carbon p_z Hamiltonian. Same-sublattice further-neighbor hopping is present. Zero on-site energy implies $\text{Tr } \mathbf{H} = 0$, but not exact particle–hole symmetry. Zero energy is not automatically the half-filling chemical potential.

For $a = 1$ and $\xi = 0.03$, the nearest-neighbor intralayer hopping is

$$t_{\text{nn}} = t_0 e^{-a/\xi} = 3.33824 \times 10^{-15} t_0. \quad (\text{S13})$$

This explains energies of order $10^{-14} t_0$ in the original plots and reciprocal DOS scales of order $10^{14}/t_0$. Such values are not floating-point underflow. They do, however, obscure the natural model units and cannot be interpreted as physical electronvolt energies without calibration. The exact dimensionless Hamiltonian is

$$\overline{H}_{ij} = \frac{H_{ij}}{t_{\text{nn}}} = \exp\left[\frac{a - |\mathbf{x}_i - \mathbf{x}_j|}{\xi}\right] \quad (i \neq j), \quad \overline{H}_{ii} = 0. \quad (\text{S14})$$

Eigenvectors are unchanged, and $\epsilon_n = E_n/t_{\text{nn}}$. This rescaling preserves all dimensionless spectral gaps and does not alter the physical content of the specified model.

The DOS must be transformed with the same change of variables. For a DOS $\rho(E)$ normalized to one state, probability conservation gives

$$\overline{\rho}(\epsilon) d\epsilon = \rho(E) dE, \quad \overline{\rho}(\epsilon) = t_{\text{nn}} \rho(t_{\text{nn}} \epsilon), \quad \overline{\eta} = \eta/t_{\text{nn}}. \quad (\text{S15})$$

Multiplying only the energy axis, while leaving the DOS and broadening unchanged, would therefore be inconsistent. The independently regenerated figures use Eqs. (S14) and (S15) throughout.

The strongest allowed interlayer matrix element is that for vertical alignment. Its ratio to the intralayer nearest-neighbor hopping is

$$t_{\perp}^{\text{max}}/t_{\text{nn}} = e^{(a-d)/\xi}. \quad (\text{S16})$$

At $\xi/a = 0.03$, the values for $d/a = 0.99, 0.98, 0.97$ are respectively 1.396, 1.948, and 2.718. This ratio changes under a change of distance and cannot be removed by a global energy rescaling. Likewise, changing ξ changes the intralayer range, for example

$$t_{\text{next}}/t_{\text{nn}} = e^{-(\sqrt{3}-1)a/\xi}. \quad (\text{S17})$$

The $\xi/a = 0.03, 0.1, 0.3$ comparison is thus a family of different Hamiltonians, not merely three choices of units. Assigning a conventional graphene hopping in electronvolts would still leave the interlayer angular dependence, distance law, sign convention, and chemical validity uncalibrated. The model's $d/a \simeq 1$ is not a prediction that real graphene should be compressed to its C–C bond length.

INDEPENDENT NUMERICAL REPRODUCTION

The reproduced model uses $R/a = 25$, giving 1510 sites in each layer and a 3020×3020 real-symmetric matrix. The earlier notebook example used a larger radius. It must not be confused with the latest parameter series. We evaluated all eigenvalues at 41 equally spaced angles from 0 to $\pi/3$, inclusive, for five parameter combinations: $(d/a, \xi/a) = (0.99, 0.03), (0.98, 0.03), (0.97, 0.03), (0.97, 0.1), (0.97, 0.3)$.

The spectra at negative angles follow exactly from those at positive angles. Exchanging the two identical layers, applying a rigid in-plane rotation by $-\theta$, and reflecting the normal coordinate maps the pair $(P_R, R_\theta P_R)$ to $(P_R, R_{-\theta} P_R)$ up to relabeling. Pair distances are unchanged, so the Hamiltonians are related by a permutation and have identical eigenvalues. This gives an 81-angle display on $[-60^\circ, 60^\circ]$. It does not establish exact 60° periodicity of the finite site-centered disk: a honeycomb lattice centered on a carbon site has a different finite-boundary symmetry from one centered on a hexagon. Nor does it justify imposing a bulk periodicity on the angle-dependent DFT patches.

The plotted model DOS is

$$\overline{\rho}_{\overline{\eta}}(\epsilon; \theta) = \frac{1}{N} \sum_{n=1}^N \frac{\exp[-(\epsilon - \epsilon_n(\theta))^2 / (2\overline{\eta}^2)]}{\sqrt{2\pi} \overline{\eta}}, \quad N = 3020. \quad (\text{S18})$$

Its integral over the entire real axis is one, not the unnormalized total number of states. Following the notebook convention, each parameter set uses the global spectral span over all sampled angles divided by 1000 as its Gaussian width. The width can therefore differ between panels. The original 500-point display grid had a spacing of about twice this width. We use 2001 points, approximately half the width, to sample the same kernel more smoothly. This

improves rendering, not the underlying physical resolution. The common color scale is clipped above $\bar{\rho} = 0.5$ to reveal weaker structures. The plotted energy window is $[-6, 6]$ in units of t_{nn} .

The source package contains `analysis/reproduce_tb.py`, `analysis/make_figures.py`, and the eigenvalue arrays in `data/tb_reproduction`. The first script constructs the normalized Hamiltonian directly and diagonalizes it with SciPy. The second computes the Gaussian DOS and assembles the figures. These are independent finite-size reproductions of the notebook model, not a new DFT calculation, a thermodynamic-limit extrapolation, or a computation of topological invariants.

WHAT WOULD ESTABLISH A TOPOLOGICAL GAP?

The acoustic experiment motivating this study combined spectral measurements with boundary-state evolution under a controllable phason [8]. Twisted mechanical bilayers supplied a more direct setting in which incommensurate geometry organizes gaps through higher-dimensional topology [9]. Neither precedent establishes topology from the appearance of a DOS image alone. A two-dimensional bilayer has two relative-translation coordinates, providing a natural phason torus. These coordinates are distinct from the twist angle. Sliding responses and bilayer gap-labeling frameworks make that distinction explicit [10–12].

For a bulk operator, the integrated DOS is defined using a trace per volume,

$$\mathcal{N}(E; \theta) = \tau [\mathbf{1}_{(-\infty, E]}(\hat{H}_\theta)]. \quad (\text{S19})$$

The normalization of τ must be fixed before a gap label is compared across geometries. An integral of lower-layer PDOS generally does not equal this trace: it measures projected spectral weight rather than the rank density of a bulk spectral projector. A finite matrix also has discrete spacings between eigenvalues, many of which need not survive as bulk gaps.

In the noncommutative-torus construction of Ref. [9], a gap's integrated density is constrained by integer coefficients and entries of a 2×2 geometric matrix A . In that work's notation and sign convention, the trace takes the form

$$\mathcal{N}(G) = n_\emptyset + n_{\{1,3\}}A_{11} + n_{\{1,4\}}A_{12} + n_{\{2,3\}}A_{21} + n_{\{2,4\}}A_{22} + n_{\{1,2,3,4\}} \det A. \quad (\text{S20})$$

The two-index and four-index integer labels refer to the four-dimensional algebra, whereas the indices of A refer to the two-dimensional lattice geometry. This expression is framework- and normalization-dependent, not a universal fitting formula for the supplied CP2K maps. Moreover, if the geometric family has $\det A = 1$, its trace constrains the constant combination $n_\emptyset + n_{\{1,2,3,4\}}$ rather than separately identifying the second-Chern coefficient. A spectral fit along that family alone therefore cannot provide a unique second-Chern assignment.

Establishing the proposed connection requires candidate gaps that survive size, boundary, reciprocal-space, and numerical convergence tests, followed by a justified gap-label or phason-response calculation for the corresponding bulk projector. No such label, Chern number, quantized pump, or physical Hall response is reported here. The real hopping Hamiltonian carries no magnetic flux. Synthetic-dimensional topology must not be conflated with the magnetic Hofstadter mechanism.

MONOLAYER REFERENCE AND REMAINING CONTROLS

The additional data contain total-DOS files for monolayer size labels 10, 20, 25, and 30 with SZV, and 10, 20, and 30 with DZVP. The labels identify the supplied size series. The accompanying description associates label 25 with approximately 1200 atoms, but monolayer coordinates and completed outputs are not included. We therefore retain the source labels rather than infer exact atom counts or reciprocal-space meshes from filenames.

Figure S2 is a common-convention replot of these files. It adds the alpha and beta DOS columns, converts the energy grid to eV, and subtracts only the Fermi energy recorded in each header. The additional, size-dependent energy shifts used in the original monolayer plotting notebook are not applied. All curves use a Gaussian of standard deviation 0.10 eV and are normalized to their own maximum in $[-7, 3]$ eV. This broader display kernel resolves the reference envelopes and is intentionally distinct from the sharper bilayer-map kernel. The curves show size-dependent discrete spectral structure and its redistribution with basis choice. They supply a monolayer reference, not a matched subtraction from the layer- and orbital-resolved bilayer observable.

A quantitative interaction-induced DOS difference would require a matched noninteracting-layer reference. With a common energy origin and identical broadening, define the unnormalized total-DOS change as

$$\Delta D(E; \theta, d) = D_{\text{bilayer}}(E; \theta, d) - D_1^{\text{ref}}(E; \theta) - D_2^{\text{ref}}(E; \theta). \quad (\text{S21})$$

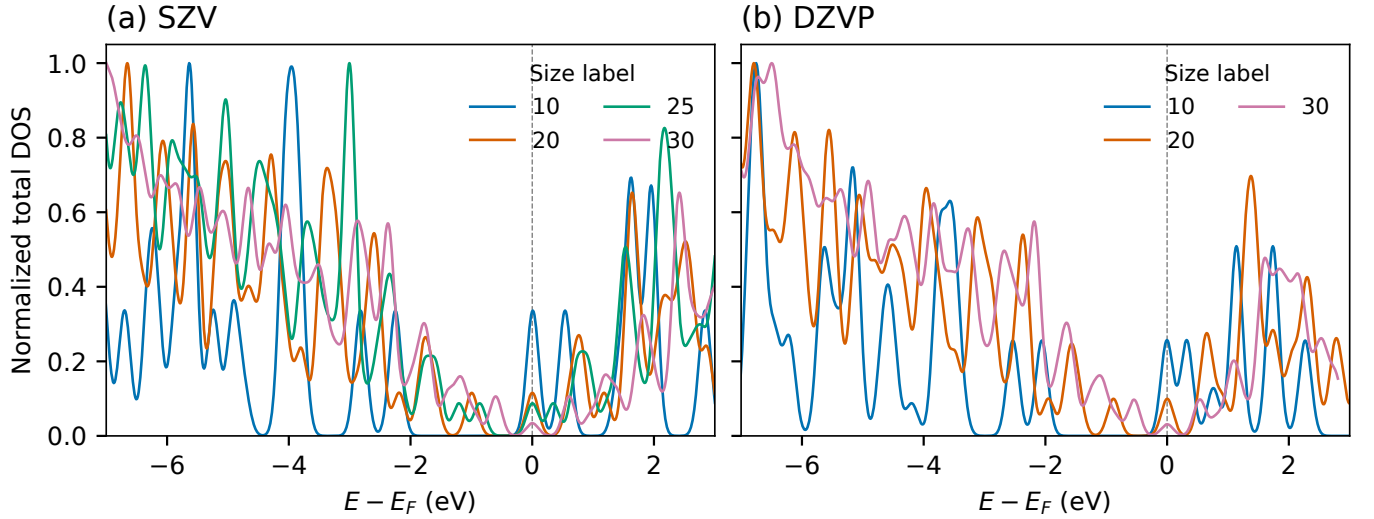


FIG. S2. Supplied monolayer total DOS for the indicated size labels with (a) SZV and (b) DZVP bases. Both spins are included. A common Gaussian standard deviation of 0.10 eV is used, with individual maximum normalization in the displayed window. Energy zero is each file's header Fermi level, without empirical offsets. Differences in discretization and the size-dependent energy reference preclude interpreting near-zero finite-cell minima as established bulk gaps.

The references retain the actual layer geometries and edge termination but remove their mutual interaction. The simplification $D_1^{\text{ref}} + D_2^{\text{ref}} = 2D_{\text{mono}}$ is valid only for equivalent layers with the same state-count normalization. For the identical model disks, if every DOS is instead normalized per state, the noninteracting reference is one monolayer DOS, not twice that DOS. In the DFT geometry the layers are inequivalent, so a periodic carbon sheet and the actual hydrogen-terminated patch must be treated separately. A consistent basis-space convention is also necessary if basis-set-superposition effects are to be separated.

Separately setting each reference's Fermi energy to zero is not a common energy alignment. A vacuum or other well-defined electrostatic reference is needed for quantitative DFT differences. When the same complete state space and normalization are used, the integral of the unnormalized ΔD over all energies vanishes: coupling redistributes states rather than creating them. A restricted occupied-state or printed-orbital window need not satisfy this sum rule. The matched-reference calculation was proposed in the earlier analysis but is not included among the supplied results.

The supplied output resolves the computational version, projection, spin selection, display convention, and finite printed-state range. Remaining convergence questions concern larger Gaussian bases and grid cutoffs, reciprocal-space sampling, vacuum thickness, upper-patch size, and boundary or coverage effects. No relaxation of the compressed structures is included in the present results. Such tests are relevant to future quantitative gap widths, pressure assignments, and topological labels. They do not change the fixed-geometry model or the spectra reported here.

COMPLETE FIRST-PRINCIPLES IMAGE SURVEY

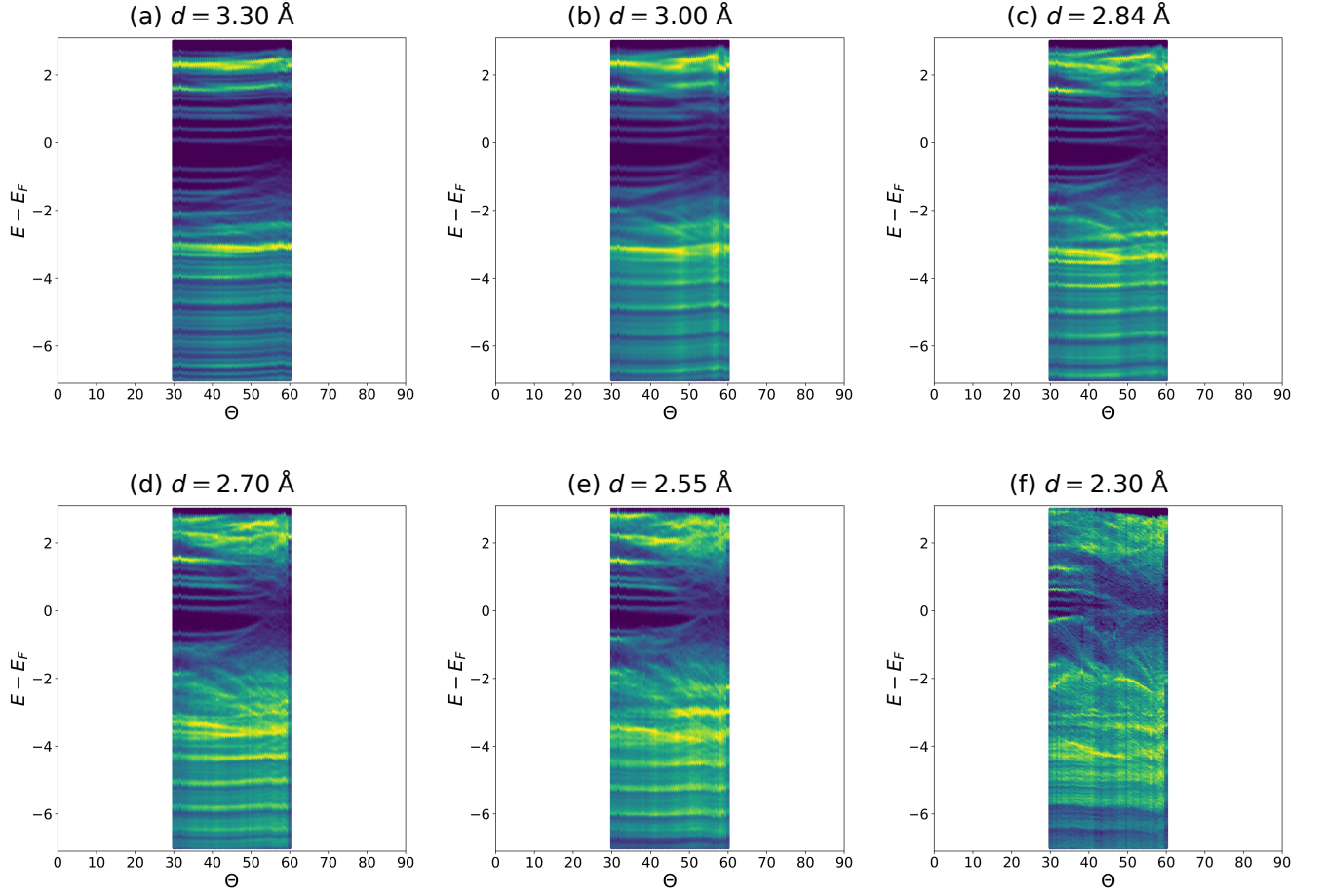


FIG. S3. Original SZV lower-layer p_z alpha-spin DOS panels for $d = 3.30, 3.00, 2.84, 2.70, 2.55$, and $2.30 \text{ \AA}$. The third separation is corrected from the source label of $2.85 \text{ \AA}$ using the coordinates. Image axes, colors, and unavailable columns are preserved. $E - E_F$ is in eV and Θ in degrees. Intensities are normalized to each angular column's maximum. The sampled interval is $30^\circ\text{--}60^\circ$.

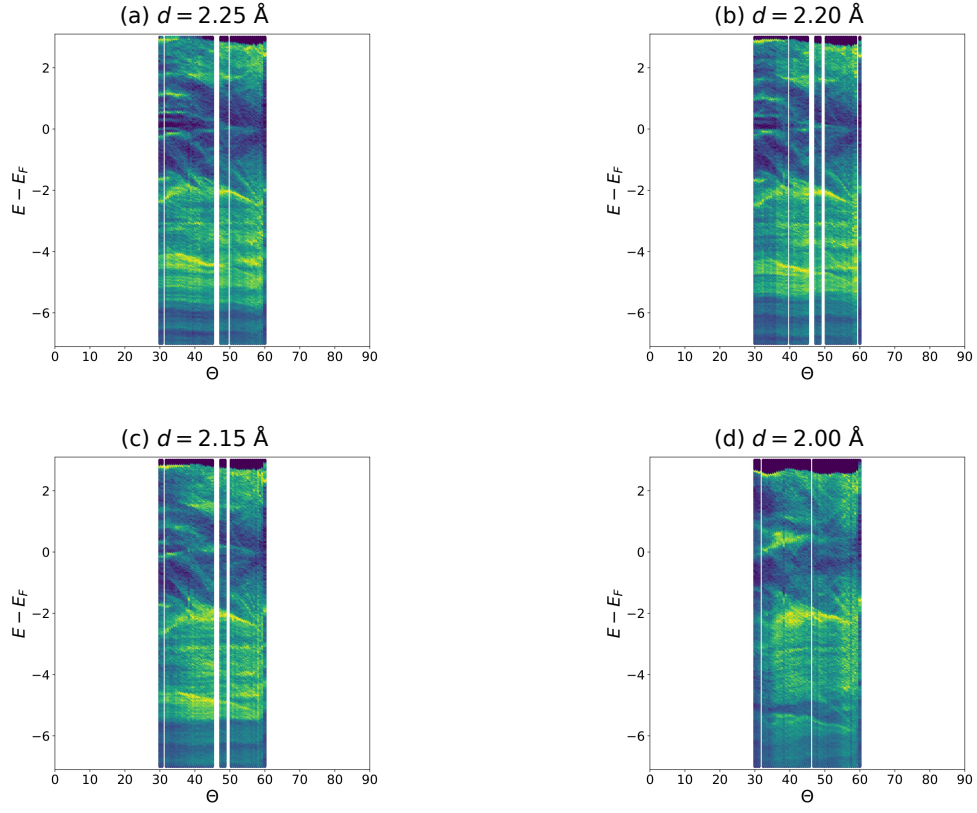


FIG. S4. Original SZV lower-layer p_z alpha-spin DOS panels for $d = 2.25, 2.20, 2.15$, and $2.00 \text{ \AA}$, with the same units and column normalization as the preceding figure. White columns are unavailable data in the original plotting workflow. State-resolved coverage and finite upper energy limits are documented in Table S2.

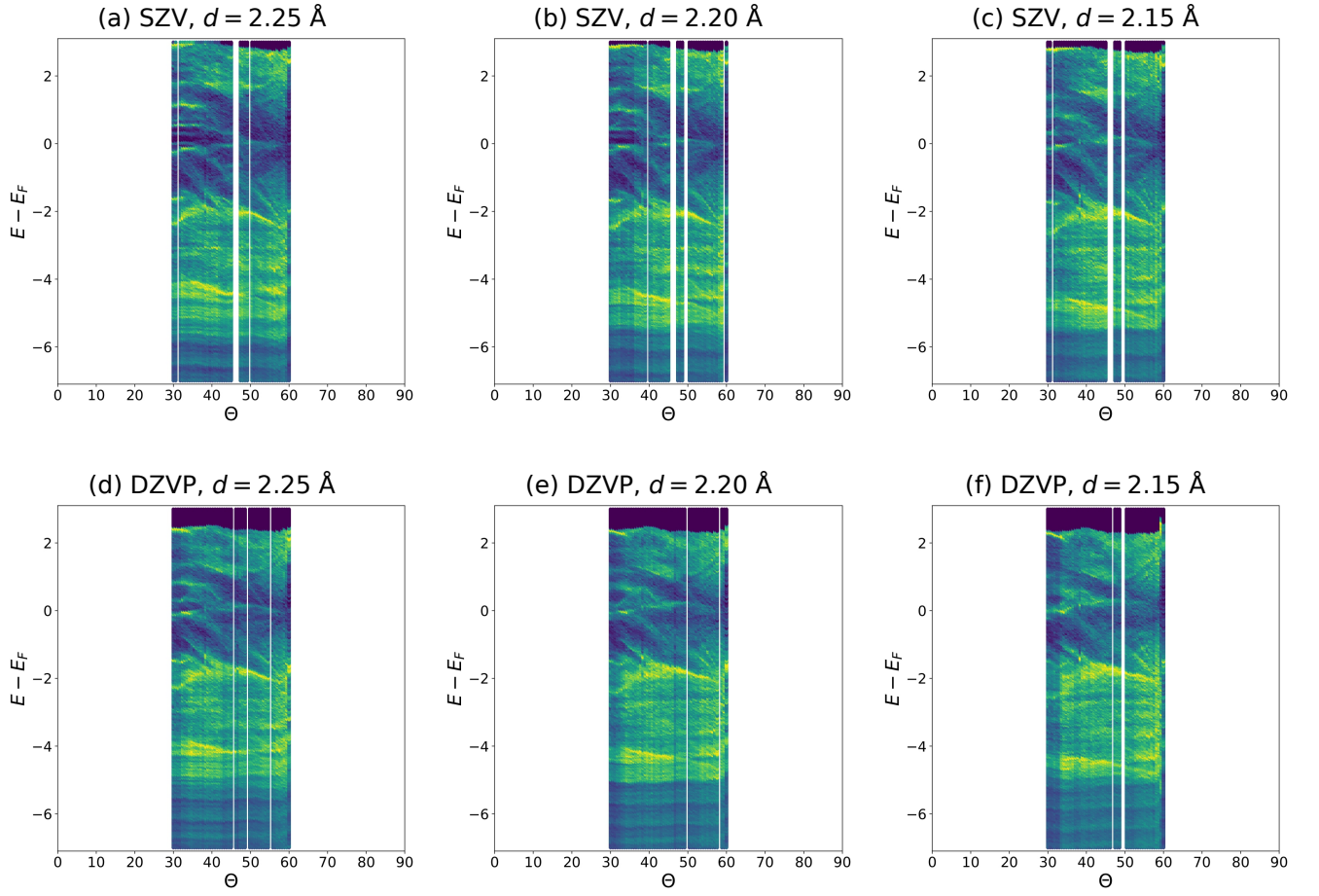


FIG. S5. Original SZV (top) and DZVP (bottom) lower-layer p_z alpha-spin panels at $d = 2.25, 2.20$, and $2.15 \text{ \AA}$. $E - E_F$ is in eV. Related large-scale structures provide a qualitative basis check under individual column normalization. The positive-energy truncation of the DZVP spectra is detailed in Table S2.

DEPENDENCE ON THE MODEL HOPPING RANGE

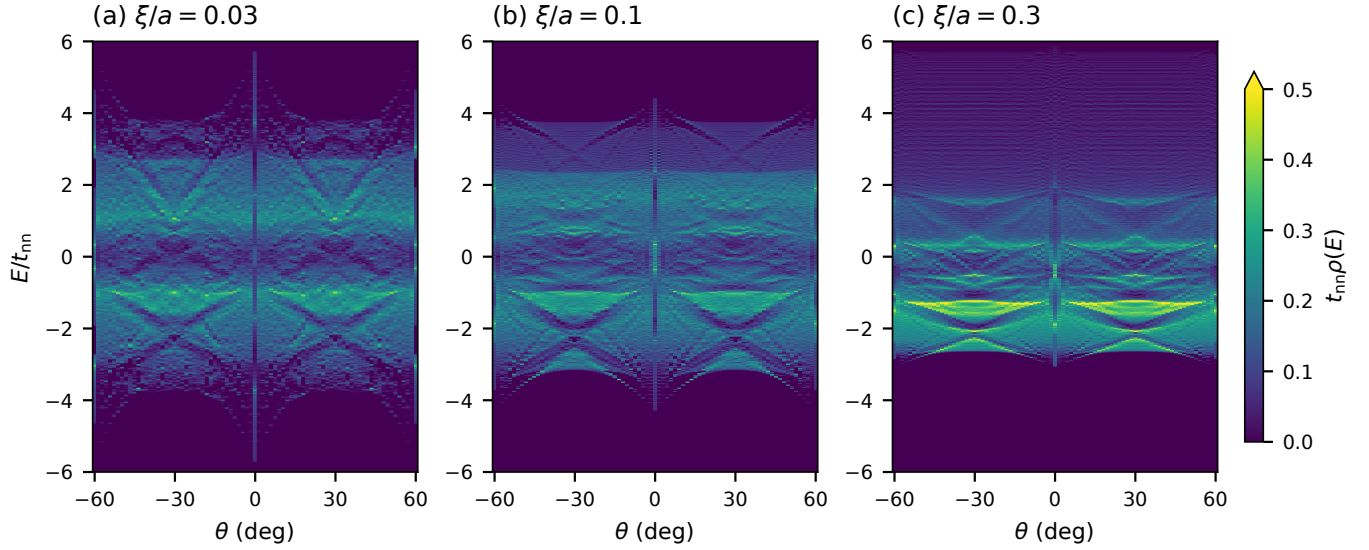


FIG. S6. Independently reproduced model DOS at fixed $R/a = 25$ and $d/a = 0.97$, for $\xi/a = 0.03, 0.1$, and 0.3 . Each Hamiltonian is normalized by its own nearest-neighbor intralayer hopping. Changing ξ changes both the relative interlayer coupling and further-neighbor intralayer hopping. Broadening follows the spectral-span convention described above. The common intensity scale is clipped above 0.5.

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